

Technology Documentation

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 | 03_Construction/Technology_Documentation.md

Official documentation links and version information for every technology, package, framework, service, and SDK used in Resilio. Organised by layer (Frontend, Backend, Database, External Services, DevOps/Tools).

FRONTEND — Flutter Mobile Application

Core Framework

Technology	Version Used	Official Documentation
Flutter SDK	3.x (stable channel)	https://docs.flutter.dev
Dart	3.x	https://dart.dev/guides
Material Design 3	Flutter built-in	https://m3.material.io

State Management and Architecture

Package	Purpose	Documentation
<code>`flutter_bloc`</code>	BLoC/Cubit state management pattern	https://bloclibrary.dev
<code>`equatable`</code>	Value equality for BLoC states/events	https://pub.dev/packages/equatable
<code>`get_it`</code>	Service locator for dependency injection	https://pub.dev/packages/get_it
<code>`injectable`</code>	Code generation annotations for GetIt	https://pub.dev/packages/injectable

Navigation

Package	Purpose	Documentation
<code>`go_router`</code>	Declarative routing with route guards	https://pub.dev/packages/go_router

Code Generation

Package	Purpose	Documentation
<code>`freezed`</code>	Immutable state models, union types, copyWith	https://pub.dev/packages/freezed
<code>`freezed_annotation`</code>	Annotations for Freezed	https://pub.dev/packages/freezed_annotation
<code>`json_serializable`</code>	JSON serialisation/deserialisation	https://pub.dev/packages/json_serializable
<code>`json_annotation`</code>	Annotations for json_serializable	https://pub.dev/packages/json_annotation
<code>`build_runner`</code>	Code generation runner	https://pub.dev/packages/build_runner

Networking

Package	Purpose	Documentation
<code>`dio`</code>	HTTP client with interceptors	https://pub.dev/packages/dio

Local Storage / Offline

Package	Purpose	Documentation
<code>`sqlite`</code>	SQLite local database	https://pub.dev/packages/sqlite
<code>`path_provider`</code>	Device file system paths	https://pub.dev/packages/path_provider
<code>`protobuf`</code>	Binary serialisation for offline sync	https://pub.dev/packages/protobuf

Authentication

Package	Version	Purpose	Documentation
`supertokens_flutter`	git master (override)	SuperTokens Flutter SDK — OTP passwordless + Google + Facebook via FDI protocol	https://supertokens.com/docs/flutter/quickstart
`google_sign_in`	latest	Google OAuth Sign-In	https://pub.dev/packages/google_sign_in
`flutter_facebook_auth`	latest	Facebook OAuth Sign-In	https://pub.dev/packages/flutter_facebook_auth

> **Note:** `supertokens_flutter` is declared under `dependency_overrides` in `pubspec.yaml` pointing to the GitHub master branch (`git: url: https://github.com/supertokens/supertokens-flutter ref: master`) because the `pub.dev` version did not support the OTP + third-party combined recipe required by Resilio.

Firestore (Multiple Services)

Package	Version	Purpose	Documentation
`firebase_auth`	^6.1.4	Firebase Authentication	https://firebase.flutter.dev/docs/auth/overview
`cloud_firestore`	^6.1.2	Cloud Firestore real-time data synchronisation	https://firebase.flutter.dev/docs/firestore/overview
`cloud_functions`	^6.0.6	Firebase Cloud Functions invocation	https://firebase.flutter.dev/docs/functions/overview
`firebase_messaging`	^16.1.1	Firebase Cloud Messaging (FCM) push notifications	https://firebase.flutter.dev/docs/messaging/overview

Real-Time Communication

Package	Purpose	Documentation
`flutter_webrtc`	WebRTC peer connection management	https://pub.dev/packages/flutter_webrtc
`socket_io_client`	Socket.IO client for signalling	https://pub.dev/packages/socket_io_client

Notifications

Package	Purpose	Documentation
<code>`firebase_messaging`</code>	Firestore Cloud Messaging (FCM)	https://firebase.flutter.dev/docs/messaging/overview
<code>`flutter_local_notifications`</code>	Local notification display	https://pub.dev/packages/flutter_local_notifications

Media and Content

Package	Purpose	Documentation	
<code>`better_player_plus`</code>	<code>^1.0.8</code>	Enhanced video player (long-form and short reels)	https://pub.dev/packages/better_player_plus
<code>`just_audio`</code>	latest	Audio playback for guided audio tracks	https://pub.dev/packages/just_audio
<code>`audioplayers`</code>	<code>^6.6.0</code>	Secondary audio playback (game sounds, UI feedback)	https://pub.dev/packages/audioplayers
<code>`cached_network_image`</code>	latest	LRU image caching from Cloudinary CDN	https://pub.dev/packages/cached_network_image
<code>`image_picker`</code>	latest	Profile photo and media selection	https://pub.dev/packages/image_picker
<code>`flutter_tts`</code>	<code>^4.2.5</code>	Text-to-speech for content narration	https://pub.dev/packages/flutter_tts

> **Note:** `video_player` and `chewie` were initially considered but replaced by `better_player_plus` which provides both long-form and short-reel (short flag) playback in a single package.

Payment

Package	Version	Purpose	Documentation
<code>`esewa_flutter_sdk`</code>	local path	eSewa payment integration	https://developer.esewa.com.np

> **Note:** `esewa_flutter_sdk` is declared as a **local path dependency** (`path: ./esewa_flutter_sdk`) in `pubspec.yaml`. The SDK is bundled directly in the project directory rather than sourced from `pub.dev`, as the `pub.dev` version did not support the eSewa v2 payment verification API required by Resilio.

Charts and Visualisation

Package	Version	Purpose	Documentation
<code>`fl_chart`</code>	latest	Mood history line/bar charts	https://pub.dev/packages/fl_chart
<code>`table_calendar`</code>	<code>^3.2.0</code>	Appointment calendar view	https://pub.dev/packages/table_calendar

UI and Animation

Package	Version	Purpose	Documentation
<code>`lottie`</code>	<code>^3.3.1</code>	Lottie animation player (loading, success states)	https://pub.dev/packages/lottie
<code>`flip_card`</code>	<code>^0.7.0</code>	Card flip animation (Wellness Trivia game)	https://pub.dev/packages/flip_card
<code>`animations`</code>	latest	Shared element and page transition animations	https://pub.dev/packages/animations

Functional Programming

Package	Version	Purpose	Documentation
<code>`dartz`</code>	<code>^0.10.1</code>	Functional Either type for error handling in domain layer	https://pub.dev/packages/dartz
<code>`rxdart`</code>	<code>^0.28.0</code>	Reactive extensions — stream composition	https://pub.dev/packages/rxdart

Utilities

Package	Version	Purpose	Documentation
<code>`intl`</code>	latest	Internationalisation, date formatting	https://pub.dev/packages/intl
<code>`flutter_localizations`</code>	built-in	I10n support for Nepali translation	Flutter built-in
<code>`shared_preferences`</code>	latest	Lightweight key-value local storage	https://pub.dev/packages/shared_preferences

Package	Version	Purpose	Documentation
`connectivity_plus`	^6.1.4	Network connectivity detection (online/offline toggle)	https://pub.dev/packages/connectivity_plus
`flutter_cache_manager`	^3.4.1	LRU content cache for offline media	https://pub.dev/packages/flutter_cache_manager
`permission_handler`	latest	Camera and microphone permission requests	https://pub.dev/packages/permission_handler
`url_launcher`	latest	Opening external URLs (Nepal crisis helpline)	https://pub.dev/packages/url_launcher

BACKEND — Node.js / Express API

Core

Technology	Version	Documentation
Node.js	20.x LTS	https://nodejs.org/en/docs/
Express.js	4.x	https://expressjs.com/en/4x/api.html

Authentication Middleware

Package	Purpose	Documentation
`supertokens-node`	SuperTokens backend SDK; session verification middleware	https://supertokens.com/docs/nodejs/quickstart

Database Client

Package	Purpose	Documentation
`@supabase/supabase-js`	Supabase JavaScript client for database operations	https://supabase.com/docs/reference/javascript/introduction

Notifications

Package	Purpose	Documentation
<code>firebase-admin</code>	Sending FCM push notifications from backend	https://firebase.google.com/docs/admin/setup

Real-Time

Package	Purpose	Documentation
<code>socket.io</code>	WebRTC signalling server	https://socket.io/docs/v4

Utilities

Package	Purpose	Documentation
<code>dotenv</code>	Environment variable management	https://www.npmjs.com/package/dotenv
<code>cors</code>	Cross-Origin Resource Sharing headers	https://www.npmjs.com/package/cors
<code>helmet</code>	HTTP security headers	https://helmetjs.github.io
<code>express-validator</code>	Request body validation middleware	https://express-validator.github.io/docs
<code>uuid</code>	UUID generation for room IDs and records	https://www.npmjs.com/package/uuid

DATABASE — Supabase (PostgreSQL)

Technology	Documentation
PostgreSQL 15	https://www.postgresql.org/docs/15/
Supabase Platform	https://supabase.com/docs
Row Level Security	https://supabase.com/docs/guides/auth/row-level-security
Supabase SQL Editor	https://supabase.com/docs/guides/database/query-files
Supabase Migrations	https://supabase.com/docs/reference/cli/supabase-db-diff

AUTHENTICATION — SuperTokens

Component	Documentation
SuperTokens Overview	https://supertokens.com/docs/community/introduction
Passwordless Recipe	https://supertokens.com/docs/passwordless/introduction
Node.js Backend SDK	https://supertokens.com/docs/nodejs/quickstart
FDI API Reference	https://supertokens.com/docs/contribute/apis/fdi
SuperTokens Managed Service	https://supertokens.com/docs/community/supertokens-core/self-hosting/overview
Google Provider Setup	https://supertokens.com/docs/thirdparty/common-customizations/sign-in-and-up/provider-config
Facebook Provider Setup	https://supertokens.com/docs/thirdparty/common-customizations/sign-in-and-up/provider-config

PUSH NOTIFICATIONS — Firebase Cloud Messaging

Component	Documentation
Firebase Console	https://console.firebase.google.com
Flutter Firebase Setup	https://firebase.flutter.dev/docs/overview
Firebase Messaging (Flutter)	https://firebase.flutter.dev/docs/messaging/overview
firebase-admin (Node.js)	https://firebase.google.com/docs/admin/setup
FCM Message Formats	https://firebase.google.com/docs/cloud-messaging/concept-options

MEDIA STORAGE — Cloudinary

Component	Documentation
Cloudinary Overview	https://cloudinary.com/documentation
Upload API	https://cloudinary.com/documentation/image_upload_api_reference
Video Upload	https://cloudinary.com/documentation/video_upload_api_reference
Unsigned Upload Presets	https://cloudinary.com/documentation/upload_presets
Transformation URL Reference	https://cloudinary.com/documentation/transformation_reference
CDN Delivery	https://cloudinary.com/documentation/advanced_url_delivery_options

REAL-TIME COMMUNICATION — WebRTC + Socket.IO

Component	Documentation
WebRTC Official	https://webrtc.org
WebRTC API Reference	https://developer.mozilla.org/en-US/docs/Web/API/WebRTC_API
flutter_webrtc	https://github.com/flutter-webrtc/flutter-webrtc
Socket.IO Server Docs	https://socket.io/docs/v4/server-api
Socket.IO Client Docs	https://socket.io/docs/v4/client-api
STUN/TURN (Google free STUN)	stun:stun.l.google.com:19302
Coturn (TURN server)	https://github.com/coturn/coturn

PAYMENT — eSewa

Component	Documentation
eSewa Developer Portal	https://developer.esewa.com.np
eSewa Flutter SDK	https://pub.dev/packages/esewa_flutter_sdk

Component	Documentation
eSewa Test Credentials	Available in developer portal after registration
Payment Flow Documentation	https://developer.esewa.com.np/#/epay

DESIGN TOOLS

Tool	Purpose	Link
Figma	High-fidelity UI mockups and prototype	https://figma.com
draw.io / diagrams.net	UML, ERD, DFD, system architecture diagrams	https://app.diagrams.net
Mermaid Live Editor	Mermaid diagram code preview	https://mermaid.live
Lucidchart	Gantt chart, WBS	https://lucidchart.com
Canva	Report visual assets	https://canva.com

DEVELOPMENT TOOLS

Tool	Purpose	Link
Android Studio	Primary Flutter IDE	https://developer.android.com/studio
VS Code	Secondary editor (backend)	https://code.visualstudio.com
Dart DevTools	Flutter widget inspector and performance profiler	https://docs.flutter.dev/tools/devtools
Postman	API endpoint testing	https://www.postman.com
DBeaver	PostgreSQL GUI for Supabase inspection	https://dbeaver.io
Git	Version control	https://git-scm.com
GitHub	Remote repository hosting	https://github.com

Tool	Purpose	Link
Android Emulator	App testing on virtual Android device	Android Studio built-in
Physical Android Device	On-device testing (WebRTC, push notifications)	—